

UK Aviation Passenger Demand Forecasting & Route Risk Analysis

Forecasting UK airport passenger demand and identifying route concentration risks to support operational resilience, workforce planning, and strategic airport investment decisions.

Executive Summary

This project analyses UK Civil Aviation Authority (CAA) passenger traffic data to evaluate airport growth patterns, route dependency risks, and passenger demand forecasting across UK domestic aviation networks.

The analysis identified significant differences in operational resilience between large hub airports, medium-growth regional airports, and small route-dependent airports.

Key findings include:

- Medium-sized airports demonstrated stronger relative recovery and growth potential compared to major UK hubs.
- 44% of analysed domestic routes exhibited high concentration risk, where a single route accounted for more than 50% of airport passenger traffic.
- Small regional airports showed significantly higher operational volatility due to charter dependency and limited route diversification.
- Forecasting models revealed divergent recovery trajectories across airport categories, highlighting the need for airport-specific planning strategies rather than a uniform national approach.
- Forecast outputs support operational planning decisions including staffing allocation, terminal capacity management, and route diversification strategy.

Business Problem

UK airports and airline operators require accurate passenger demand forecasting to support:

- Capacity planning
- Workforce optimisation
- Route network management
- Operational resilience
- Revenue stability

Passenger demand recovery following market disruptions has been uneven across the UK aviation sector. While some airports have demonstrated strong recovery trajectories, others remain operationally vulnerable due to high dependency on limited domestic routes or charter traffic.

This project evaluates airport-level passenger demand trends, identifies structural concentration risks, and develops forecasting models to support strategic and operational decision-making.

Business Objectives

The project aimed to:

- Analyse passenger growth trends across UK airports
- Segment airports by operational scale and growth profile.
- Assess scheduled versus charter traffic dependency.
- Identify route concentration risks across regional airports.
- Forecast airport and route-level passenger demand over a 3–5 years horizon.
- Provide operational and strategic recommendations for airport planning teams.

Stakeholders

This analysis is relevant to:

- Airport Operations Teams
- Airline Network Planning Departments
- Commercial Strategy Teams
- Ground Operations Management
- Workforce Planning Teams
- Aviation Infrastructure & Investment Teams

Data Source

Source:

UK Civil Aviation Authority (CAA) passenger traffic datasets

Dataset Scope:

- UK domestic passenger traffic
- Airport-level passenger volumes
- Route-level passenger demand
- Scheduled and charter traffic segmentation.
- Five-year historical comparison period

Coverage:

- 40+ UK airports
- Multiple domestic route combinations

- Historical passenger demand and growth metrics

Analytical Methodology

1. Data Cleaning & Preparation

Passenger datasets were cleaned and standardised using Python. Airport naming inconsistencies, missing values, and route-level formatting issues were resolved prior to analysis.

Key preprocessing tasks included:

- Airport classification
- Passenger aggregation
- Route normalisation
- Time-series preparation
- Growth metric calculations

2. Airport Segmentation

Airports were segmented into:

- Large
- Medium
- Small

Segmentation was based on total passenger volume and operational scale.

The analysis evaluated:

- Passenger recovery trends
- Compound Annual Growth Rate (CAGR)
- Demand volatility
- Relative growth performance

3. Traffic Composition Analysis

Scheduled and charter passenger shares were calculated to assess operational stability and volatility exposure.

This analysis identified:

- Stable hub-airport demand structures
- Charter-heavy regional airport dependency
- Airports vulnerable to seasonal demand fluctuations

4. Route Dependency Analysis

Route concentration metrics were developed to evaluate operational dependency risk.

Airports where a single route represented more than 50% of total passenger demand were flagged as high-risk.

This framework helped identify:

- Structurally fragile regional airports
- Revenue concentration exposure
- Operational vulnerability to route cancellations

5. Forecasting & Demand Modelling

Passenger demand forecasts were developed using:

- ARIMA models
- Prophet forecasting models

Forecasting was performed at:

- Airport level
- Route level

Models incorporated:

- Trend behaviour
- Seasonality
- Demand variability
- Recovery trajectory analysis

6. Model Validation

Forecast accuracy was evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)

Prophet models demonstrated stronger performance in volatile regional airport demand scenarios, while ARIMA models performed effectively for stable long-term trend forecasting.

Key Findings

1. Medium-Sized Airports Show Strongest Growth Potential

The analysis identified medium-sized airports such as:

- Luton

- Edinburgh
- Birmingham

as demonstrating stronger relative recovery and growth potential compared to major UK hub airports.

Large airports maintained high passenger volumes but exhibited slower long-term growth trajectories.

Business Implication

Targeted infrastructure investment in medium-growth airports may generate stronger marginal returns than expansion at saturated hub airports.

2. Small Regional Airports Are Highly Vulnerable

Several small regional airports were found to rely heavily on a single domestic route.

Examples included:

- Barra
- Tiree
- Campbeltown
- Wick John O’Groats

In some cases, a single route accounted for more than 95% of total airport passenger demand.

Business Implication

Route cancellations or demand shocks could significantly impact operational viability and airport revenue stability.

3. Charter Dependency Increases Operational Volatility

Large airports operated almost entirely on scheduled traffic models, while several smaller airports exhibited high charter dependency.

Examples:

- Biggin Hill
- Southend
- Prestwick

Business Implication

Charter-heavy airports require flexible operational and staffing models to manage unpredictable passenger demand patterns.

4. Route Concentration Risk Is Widespread

44% of analysed routes were classified as high dependency.

Several airports demonstrated excessive reliance on:

- Heathrow domestic connectivity
- Glasgow regional links
- Aberdeen regional routes

Business Implication

Network diversification is critical for improving resilience and reducing operational fragility.

5. Forecasting Revealed Divergent Recovery Patterns

Passenger recovery trajectories differed significantly by airport category.

Findings showed:

- Strong CAGR growth among selected regional airports
- Continued stagnation at some major hubs
- High volatility in smaller regional markets

Business Implication

Airport planning strategies should be tailored to local demand conditions rather than relying on uniform national forecasting assumptions.

Forecasting Results

Forecast models projected continued uneven recovery patterns across the UK domestic aviation network.

Key forecast outcomes included:

- Continued passenger growth at medium-growth airports
- Persistent concentration risk at regional airports
- Increased vulnerability among highly specialised route-dependent airports

Forecast outputs support:

- Workforce planning
- Seasonal staffing decisions
- Capacity allocation
- Infrastructure planning
- Route diversification strategy

Operational Recommendations

1. Diversify High-Risk Route Networks

Airports with excessive dependency on single domestic routes should prioritise:

- Airline partnership expansion
- Route diversification initiatives
- Demand balancing strategies

2. Prioritise Investment in Medium-Growth Airports

Medium-sized airports demonstrating strong recovery trajectories should be prioritised for:

- Capacity expansion
- Terminal investment
- Operational scaling

3. Implement Flexible Operating Models

Charter-heavy and highly seasonal airports should adopt:

- Dynamic staffing models
- Flexible operational planning
- Scalable workforce allocation

4. Embed Forecasting into Planning Cycles

Passenger forecasting models should support:

- Budget forecasting
- Workforce planning
- Seasonal scheduling
- Operational risk management

Assumptions & Limitations

- Forecasts assume relatively stable macroeconomic conditions.
- Passenger-level fare and revenue data were unavailable.
- COVID-era demand shocks may distort long-term forecasting behaviour.
- Route-level granularity varies across smaller regional airports.

Future Improvements

Potential future enhancements include:

- Real-time flight operations integration
- Weather disruption modelling
- Revenue and yield forecasting
- Airline-level segmentation
- Machine learning forecasting approaches
- Interactive dashboard deployment